

Project Documentation

Waste-to-Value

B2B Contract-Based Waste Logistics & Commodity Trading Platform

Prepared for DEPI Project Documentation

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Prepared By

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DEPI Documentation Milestones

The following milestone table follows the DEPI documentation guideline structure and confirms that the document covers the required academic deliverables.

Item	Student Deadline	Graduate Deadline
Project Planning & Management	2/20/2026	1/27/2026
Literature Review	4/20/2026	3/1/2026
Requirements Gathering	4/20/2026	3/1/2026

System Analysis & Design	5/1/2026	4/3/2026
Implementation (Source Code & Execution)	7/10/2026	5/17/2026
Final Presentation, Testing & Reports	7/17/2026	5/24/2026

1. Project Planning & Management

1.1 Project Proposal

Waste-to-Value is a compliance-first B2B contract-based waste logistics and commodity trading platform. It acquires pre-segregated recyclable materials from corporate waste generators, processes them through a controlled aggregation hub, and delivers mill-ready baled commodities to industrial recycling factories.

The project transforms corporate waste from an operational liability into a traceable economic asset by connecting shopping malls, hotels, corporate headquarters, universities, and sports clubs with industrial off-takers through a structured digital workflow.

1.2 Project Overview

Element	Description
Project Name	Waste-to-Value
Business Type	B2B contract-based waste logistics and commodity trading
Main Users	Supplier, Factory, Driver, Hub Manager, Super Admin
Core Pipeline	Source Extraction, Hub Inbound & QA, Densification & Inventory, Factory Offtake

Pricing Rule	Buy at 70% of market rate and sell at 110% of market rate
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1.3 Objectives

- Automate buy and sell pricing using a zero-bidding Algorithmic Pricing Engine.
- Enforce a strict 5% contamination threshold during hub inspection.
- Provide role-based access control for operational and financial data.
- Deliver live analytics for tonnage, stock levels, fleet utilization, and margin performance.
- Coordinate inbound dry-leased collection routes and outbound 24-ton FTL factory deliveries.
- Support onboarding and FAQs using a restricted Gemini AI chatbot.
- Build a scalable architecture that can expand to multiple hubs and governorates.

1.4 Scope

In Scope	Out of Scope
Responsive role-based web dashboards	No consumer-facing household mobile app
Supplier and factory onboarding workflows	No open bidding marketplace
Hub QA, weighing, contamination and inventory tracking	No truck ownership; inbound dry lease and outbound subcontracting
Pricing engine, ESG page, and operational reports	No in-system payment gateway; bank transfer remains outside the platform

1.5 Task Assignment & Roles

The first-page table identifies the official team position. The detailed work distribution below is used inside the documentation to explain implementation responsibilities.

Team Member	Detailed Responsibility
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Shahd Saber Abdelmohsen	Team Leader; Front-End Development + Back-End Development
Mohamed Adel Ibrahim Mohamed	Front-End Development
Norhan Mohamed Bakr Ahmed	Front-End Development + System Analysis
Atef Mustafa Essa	Front-End Development + Documentation

1.6 Risk Assessment & Mitigation Plan

Risk	Impact	Mitigation
Factory payment delay	Cash-flow pressure and delayed supplier settlement	Apply Net 14/30 terms, late penalties, and invoice status monitoring.
High contamination ratio	Rejected loads or reduced accepted weight	Apply two-stage weighing, QA sorting, and the 5% threshold rule.
Baler downtime	Hub throughput disruption	Schedule preventive maintenance and maintain daily stock visibility.
Low supplier onboarding	Failure to reach target tonnage	Use B2B sales demos, ESG positioning, and pilot contracts.



Unauthorized access	Exposure of operational or financial data	Use authentication, role permissions, validation, and audit logs.
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1.7 KPIs

KPI	Target / Measurement
Daily material acquisition	5 tons/day operating target
Monthly throughput	130 to 150 tons/month depending on operating days
Contamination compliance	Loads should remain at or below the 5% contamination threshold
Shipment readiness	Commodity stock reaches 24-ton FTL threshold
System performance	Fast dashboard response and reliable workflow updates
User adoption	Active use by suppliers, factories, drivers, hub managers, and admins

1.8 Project Methodology

The project follows the Waterfall methodology because the documentation, analysis, design, implementation, testing, and final reporting phases are clearly sequenced.



1. **Requirements Gathering & Analysis:** Defined to support the B2B marketplace and three primary user roles.
2. **System Design:** Architectural design of mobile-first and high-load web dashboards using ERD, Use Case, Sequence, and Activity diagrams.
3. **Implementation (Coding):** Coding with React (frontend) and PHP/Laravel (backend), integrating Gemini AI.
4. **Testing:** Cross-browser (Chrome, Safari, Edge) and mobile (3G/4G) testing of all core functionalities.
5. **Deployment:** Secure cloud hosting (AWS) and initial rollout in Greater Cairo and Alexandria.
6. **Maintenance:** Ongoing support, dispute management, and planning for future expansions.

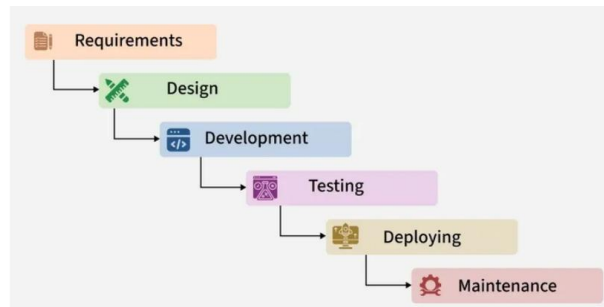


Figure 1.1: Waterfall Methodology Stages

1.9 Project Timeline

Figure 1.1: Waterfall Methodology Stages

1.9 Project Timeline

Phase	Main Deliverables	Target Period
Phase 1	Initial documentation, business plan, requirements, use cases, ERD and mapping	Feb - Mar 2026
Phase 2	20% implementation, database setup, core UI, activity and sequence diagrams	Mar - Apr 2026
Phase 3	40% implementation and backend logic	Apr 2026
Phase 4	100% implementation, final diagrams, class diagram	Apr - May 2026
Phase 5	Final documentation, testing, reports, presentation, and submission preparation	May - Jun 2026

Figure 1.2: Project Timeline Table - Part 1

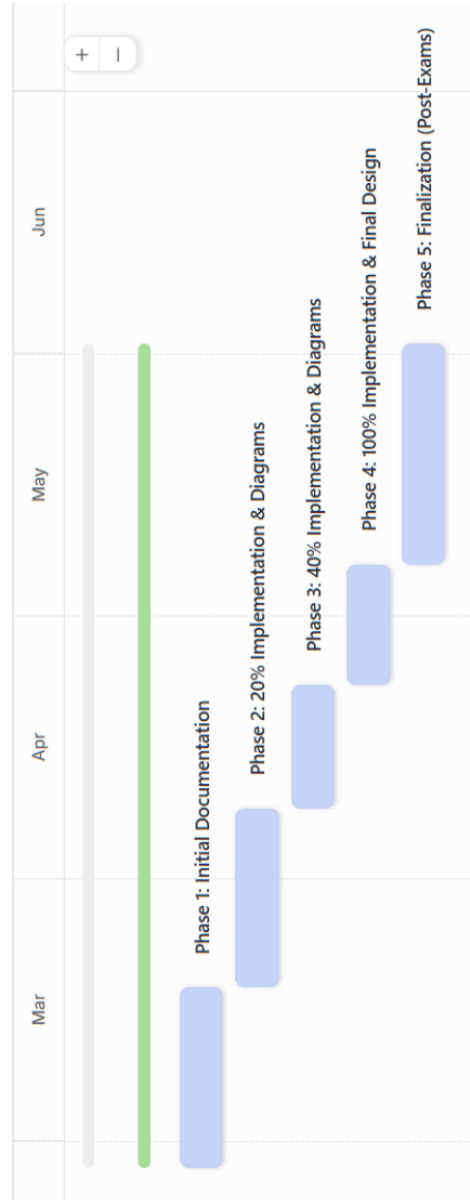


Figure 1.2: Gantt Chart for All Phases

Figure 1.3: Gantt Chart for All Phases

2. Literature Review

2.1 Background

Waste management and recycling platforms are increasingly important because corporate waste streams require traceability, quality assurance, and sustainable processing. Existing B2C recycling solutions often depend on small household volumes, reward points, or manual communication, which makes volumes unpredictable and compliance reporting weak.

Waste-to-Value takes a different approach by targeting corporate waste generators through fixed-term contracts. This creates predictable bulk supply, enables formal ESG documentation, and provides factories with a higher-quality upstream stream of baled commodities.

2.2 Competitor Differentiation

Feature	Waste-to-Value	Typical B2C Platforms
Supply Source	Corporate generators such as malls, hotels, HQs, universities, and clubs	Households and small informal sources
Business Model	B2B contract-based logistics and commodity trading	B2C collector or middleman model
Pricing	Algorithmic buy 70% / sell 110%	Points, rewards, or case-by-case negotiation
Compliance	ESG and formal reporting focus	Limited formal compliance support
Infrastructure	Own 500 sqm hub, 50-ton baler, and industrial scale	Often no owned processing hub

2.3 Suggested Improvements

- Add GPS route tracking for driver dashboards in later phases.



- Add barcode or QR scanning for bale-level traceability.
- Integrate automated notifications for supplier pickups, factory delivery dates, and payment status.
- Expand the analytics dashboard to support forecasting and profitability analysis by commodity.

2.4 Evaluation Criteria

Criteria	Evaluation Focus
Documentation	Completeness, academic structure, diagrams, and clarity
Implementation	Functional dashboards, backend APIs, database, and business rules
Testing	Test cases, validation, error handling, and user acceptance testing
Presentation	Clear explanation of the problem, solution, workflow, and final results



Figure 2.1: SDG Goals

2.3.3 SWOT Analysis

Table 2.3: SWOT Analysis for Strengths & Weaknesses

Strengths	Weaknesses
First B2B contract-based waste logistics platform in Egypt targeting corporate generators directly, not households.	Heavy upfront CAPEX requirement (558,250 EGP for hub equipment and tech alone) before generating first revenue.
Algorithmic Pricing Engine eliminates bidding entirely; guaranteed 40-point spread (buy at 70%, sell at 110%) locks in structural margins every ton.	Dependency on factory payment discipline - Net 14 terms mean 14+ days of float risk on every FTL delivery.
MoE Memorandum of Understanding provides regulatory credibility and enables formal ESG compliance reporting for corporate clients.	Small founding team (6 members) managing 13 operational staff across hub, fleet, and sales.
Own processing infrastructure (500sqm hub, 50-ton baler, 1-ton scale) ensures quality control and value-added densification.	Single hub creates geographic concentration risk; any baler downtime could halt the entire throughput pipeline.
Asset-light split logistics model (dry-lease inbound, per-trip subcontracted outbound) minimizes capital risk.	Platform requires consistent 5T/day intake to cover fixed costs; underperformance in source onboarding directly erodes margins.

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Figure 2.1: SDG Goals and SWOT Snapshot

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs
Supplier / Corporate Waste Generator	Scheduled pickups, digital weight receipts, payout history, ESG information, and support.
Factory / Recycling Buyer	Available baled inventory, delivery dates, invoices, due dates, and receipt confirmation.
Driver	Daily routes, assigned truck, pickup confirmation, photo upload, and hub-arrival updates.
Hub Manager	Inbound weighing, QA results, contamination calculation, inventory update, and baling supervision.
Super Admin	User management, contract approvals, baseline pricing, analytics, delivery notes, and invoice status updates.

3.2 User Stories

User	User Story
Supplier	As a supplier, I want to view upcoming pickups and digital receipts so that I can track my contracted waste

	deliveries.
Factory	As a factory, I want to view available baled inventory and invoice due dates so that I can manage procurement.
Driver	As a driver, I want to view daily routes and update load status so that the hub receives accurate arrival information.
Hub Manager	As a hub manager, I want to record weights and contamination so that accepted material is calculated correctly.
Super Admin	As a super admin, I want to manage contracts and pricing baselines so that the platform operates under approved rules.

3.3 Functional Requirements

- The system shall authenticate users and display role-based dashboards.
- The system shall allow suppliers and factories to submit registration requests.
- The system shall allow the Super Admin to review requests, register contracts, and create pickup schedules.
- The system shall allow drivers to view routes, confirm arrival, upload photos, and update transit status.
- The system shall calculate contamination ratio, accepted weight, and supplier invoice values.
- The system shall update commodity inventory and alert at the 24-ton shipment threshold.
- The system shall generate factory sales invoices at 110% of baseline market price.
- The system shall provide an AI chatbot for onboarding and support only.

3.4 Non-Functional Requirements

Category	Requirement
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Performance	Dashboards should load quickly and update operational data without noticeable delay.
Security	Role-based authorization must protect sensitive operational and financial data.
Usability	Forms, tables, notifications, and dashboards should be clear and responsive.
Reliability	Critical operational records such as weights, receipts, and invoices must be stored safely.
Scalability	Architecture should support future hubs, routes, factories, and governorates.

4. System Analysis & Design

4.1 Problem Statement

Corporate recyclable waste in Greater Cairo is often diverted to informal intermediaries or landfill without consistent pricing, traceability, quality assurance, or ESG documentation. Waste-to-Value solves this issue by formalizing the supply chain through contracts, scheduled logistics, controlled hub processing, and factory offtake.

4.2 System Users

User Role	Main Permissions
Supplier	View profile, pickups, weight receipts, payout history, ESG information, and chatbot support.
Factory	View inventory, estimated delivery, sales invoices, due dates, and confirm receipt.
Driver	View daily routes, confirm supplier arrival, upload load photo, and confirm hub arrival.
Hub Manager	Confirm hub arrival, record Tier 1/Tier 2 weights, supervise QA, inventory, and baling.
Super Admin	Approve onboarding, manage contracts, update market prices, view analytics, and update invoices.

4.3 Use Case Diagrams

The use case model defines the functional boundaries of the platform and describes how each role interacts with the system.



3.2 Use Case Diagrams

Use cases are the primary drivers for all UML diagramming techniques, communicating at a high level what the system needs to do. A use case diagram illustrates in a very simple way the main functions of the system and the different kinds of users who will interact with it [30]. Use Case Diagrams therefore define the functional boundaries of the system by capturing all interactions between external actors and the platform's use cases. The Waste-to-Value system involves five actors - Supplier, Factory, Driver, Hub Manager, and Super Admin - plus a generic "Any User" actor for shared functionality. The diagrams are presented in a decomposed format: first a consolidated overview, then per-actor breakdowns for clarity.

3.2.1 Use Case Diagram for All Users (Overview)

Figure 3.1 presents the consolidated Use Case Diagram for all actors within the system. It illustrates the full scope of the platform's functional requirements across all five user roles, including shared use cases (Sign In / Register and Chat with AI) that are available to every authenticated user.

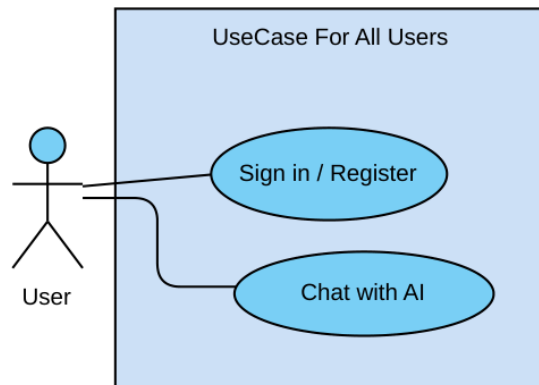


Figure 3.1: Use Case Diagram for All Users

Figure 4.1: Use Case Diagram for All Users



3.2.2 Use Case Diagram for Supplier and Factory

Figure 3.2 details the use cases specific to the **Supplier** (corporate waste generator) and the **Factory** (recycling buyer). Suppliers can submit a registration request, view their profile, check upcoming pickup schedules, and review weight receipts and payout history. Factories can view their sales invoices and due dates, check estimated delivery timelines, and confirm receipt of shipments.

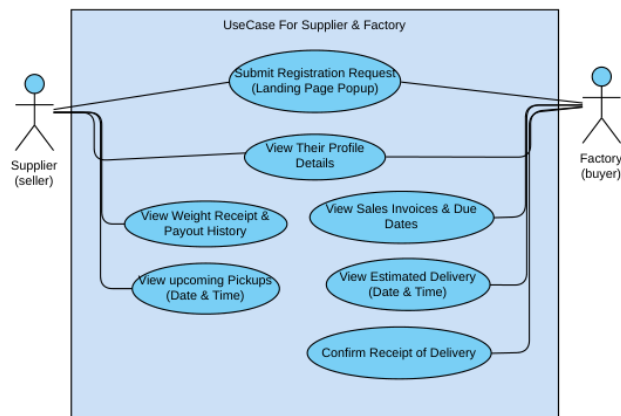


Figure 3.2: Use Case Diagram for Supplier and Factory

3.2.3 Use Case Diagram for Hub Manager

Figure 3.3 presents the detailed use cases for the **Hub Manager**, who is responsible for the entire hub processing pipeline. The Hub Manager confirms truck arrivals, performs two-stage weighing (Tier 1 and Tier 2), supervises sorting and decontamination by QA Sorters, manages inventory updates, supervises the baling process, and monitors stock levels to report daily to the Super Admin.

Figure 4.2: Use Case Diagram for Supplier and Factory

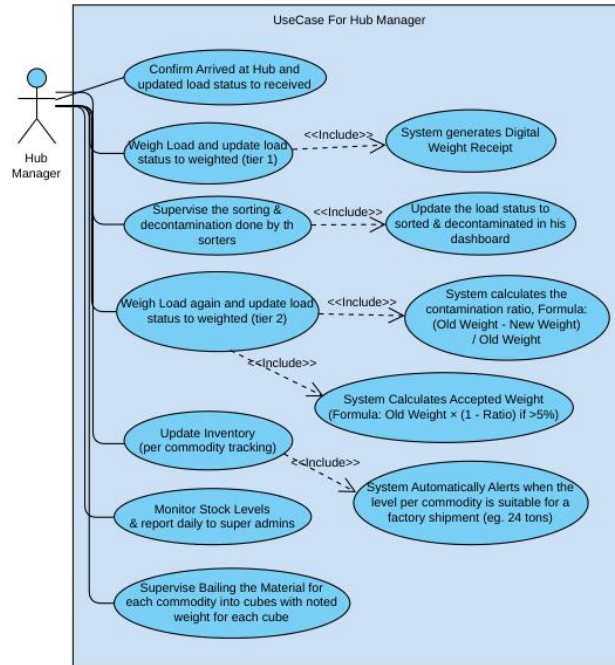


Figure 3.3: Use Case Diagram for Hub Manager

3.2.4 Use Case Diagram for Super Admin and Driver

Figure 3.4 illustrates the use cases for the **Super Admin** and the **Driver / Loader**. The Super Admin manages the registration workflow (reviewing requests, scheduling meetings, registering contracts, creating pickup schedules) and periodically updates the base market price fed into the Algorithmic Pricing Engine. The Driver views daily pickup routes, confirms arrivals at supplier locations, logs load details and photos, and confirms arrival at the hub.

Figure 4.3: Use Case Diagram for Hub Manager

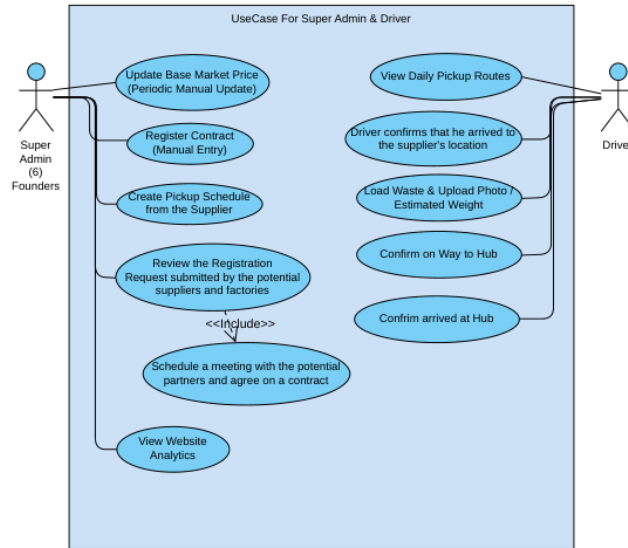


Figure 3.4: Use Case Diagram for Super Admin and Driver

3.3 Activity Diagrams

Logical process models describe processes without suggesting how they are conducted, allowing analysts to focus on how the business should run without being distracted by implementation details [31]. Activity diagrams serve this purpose in the UML by modeling the sequential and conditional flow of operations. The Waste-to-Value activity model is therefore decomposed into five parts, each representing a distinct operational phase. This decomposition ensures clarity and academic traceability across the four pipeline stages.

3.3.1 Activity Diagram Part 1 - Registration and Contract Flow

Figure 3.5 describes the onboarding workflow. A Supplier or Factory submits a registration request through the platform's landing page popup. The Super Admin then reviews the request, schedules an offline meeting to agree on contractual terms, manually registers

Figure 4.4: Use Case Diagram for Super Admin and Driver

4.4 Activity Diagrams

Activity diagrams show the sequential business behavior across registration, inbound logistics, hub processing, outbound delivery, and common user activities.



Waste-to-Value - Activity Diagram
Part 3: Hub Manager Detailed Processing (Stage 2 & 3)

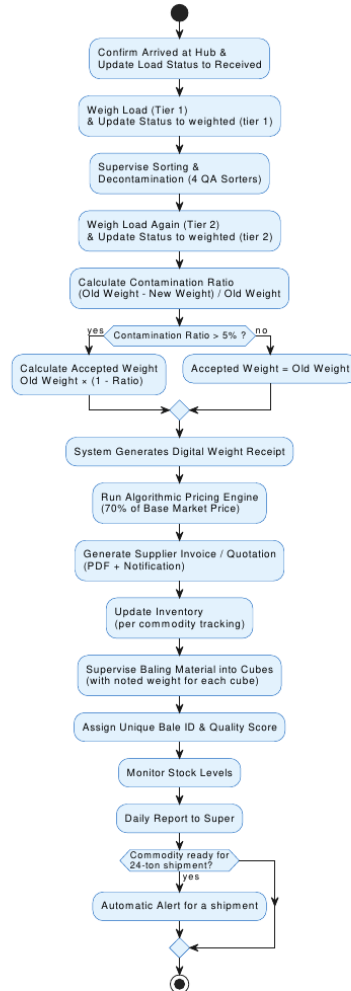


Figure 3.7: Activity Diagram Part 3 - Hub Manager Detailed Processing (Stages 2 and 3)

Figure 4.5: Activity Diagram - Hub Manager Processing Stages 2 and 3



Waste-to-Value - Activity Diagram
Part 4: Outbound & Factory Confirmation (Stage 4)

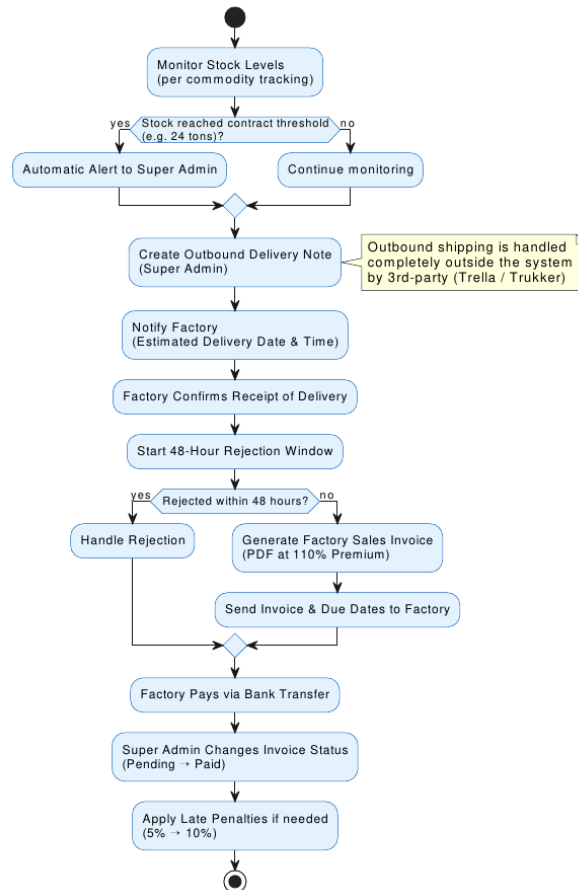


Figure 3.8: Activity Diagram Part 4 - Outbound and Factory Confirmation (Stage 4)

Figure 4.6: Activity Diagram - Outbound and Factory Confirmation Stage 4

4.5 Sequence Diagrams

Sequence diagrams describe the messages passed between the actors and the system over time for the most important operational scenarios.

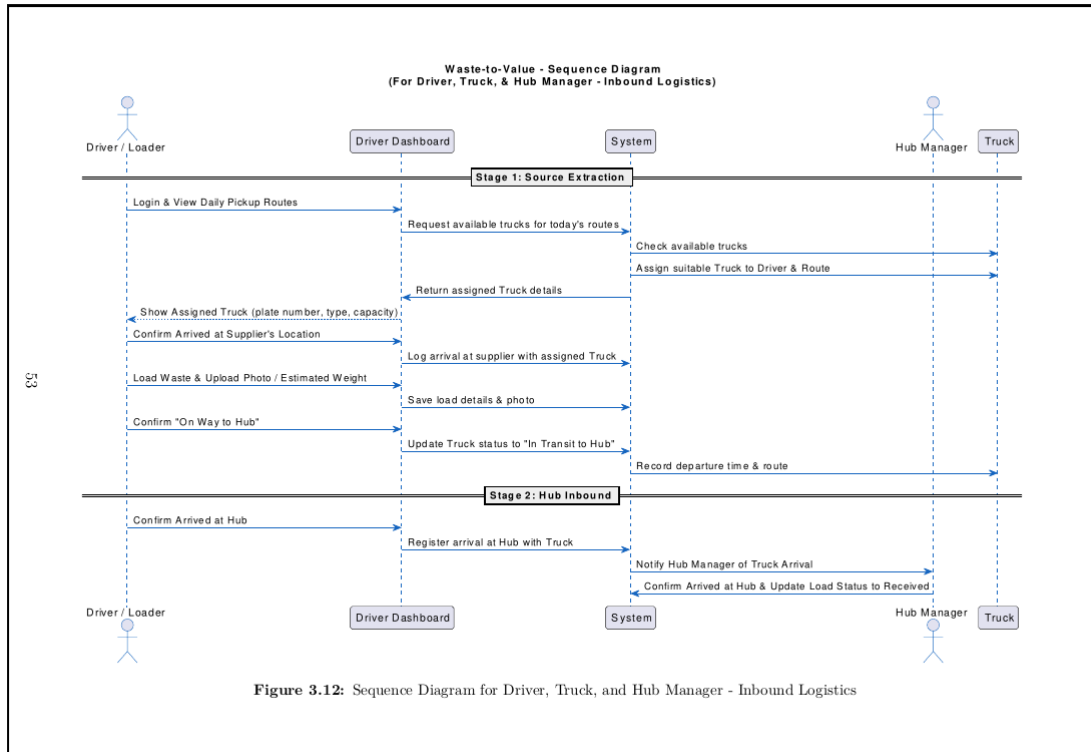


Figure 4.7: Sequence Diagram - Inbound Logistics

Figure 4.9: Sequence Diagram - Factory and Super Admin Outbound & Payment

4.6 System Flow Summary

Stage	Actors	Key System Actions	Output / Trigger
1	Driver	View routes, confirm supplier arrival, upload photo and estimated weight, confirm hub arrival	Hub Manager notified
2	Hub Manager / QA Sorters	Tier 1 weigh, sort, decontaminate, Tier 2 weigh, calculate accepted weight	Digital weight receipt and supplier invoice at 70%
3	Hub Manager	Update inventory, supervise baling, assign Bale IDs, monitor stock	Daily report and 24-ton alert
4	Super Admin / Factory	Create delivery note, notify factory, confirm receipt, manage rejection window	Sales invoice at 110% and payment status update

4.7 Class Diagram and Enumerations

The class design is centered on a base User class specialized into SuperAdmin, HubManager, Driver, Supplier, and Factory. Supporting classes model contracts, pickups, trucks, inbound records, bale cubes, commodities, pricing baselines, outbound deliveries, invoices, and chatbot support.



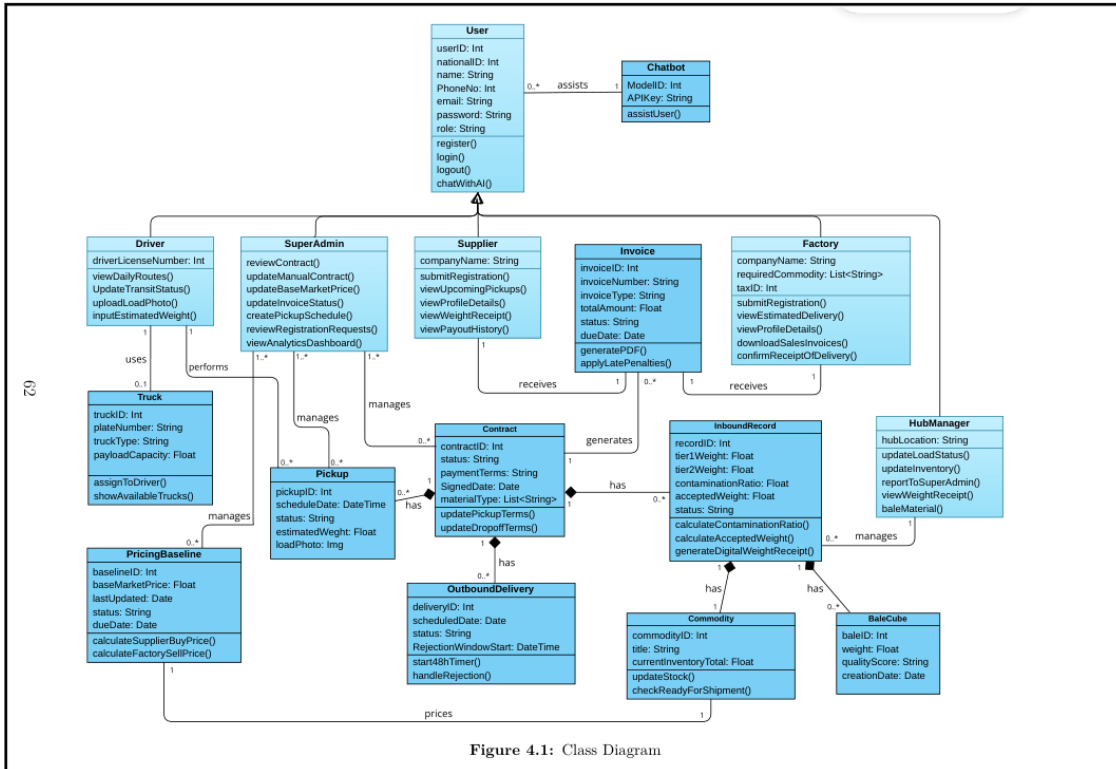


Figure 4.1: Class Diagram

Figure 4.10: Class Diagram

4.3 Database Structure (ERD)

The Entity-Relationship Diagram (ERD) is a graphical representation of the Waste-to-Value system's database schema. It visually depicts the entities in the system and how those entities are related to one another, providing a logical guide for how information is stored and retrieval. A well-designed ERD promotes data integrity and facilitates efficient information retrieval [34]. The main components represented by the ERD are as follows:

- Entities.** Each entity corresponds to a database table and represents a distinct type of object in the system. The primary entities in the Waste-to-Value database are:
- User** - Stores identity and authentication data for all platform users. Specialised into **SuperAdmin**, **HubManager**, **Driver**, **Supplier**, and **Factory** sub-entities.
 - Contract** - Records the signed agreements between the Super Admin and Suppliers or Factories, including material types, payment terms, and signed date.
 - Pickup** - Represents a scheduled waste collection event, linked to a Driver, a Supplier's Contract, and a Truck.
 - Truck** - Stores the physical vehicle details used in each pickup.
 - InboundRecord** - Captures the two-stage weighing and QA results for each inbound truck load at the hub.

Figure 4.2: Class Diagram - Enumeration Types

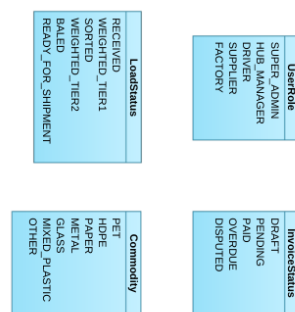


Figure 4.11: Class Diagram Enumeration Types

4.8 ERD and Database Design

The database design stores users, role-specific user data, contracts, pickups, trucks, inbound records, bale cubes, commodities, pricing baselines, outbound deliveries, invoices, and chatbot configuration. Foreign keys preserve the operational flow from contract to pickup, hub record, bale inventory, delivery, and invoice.

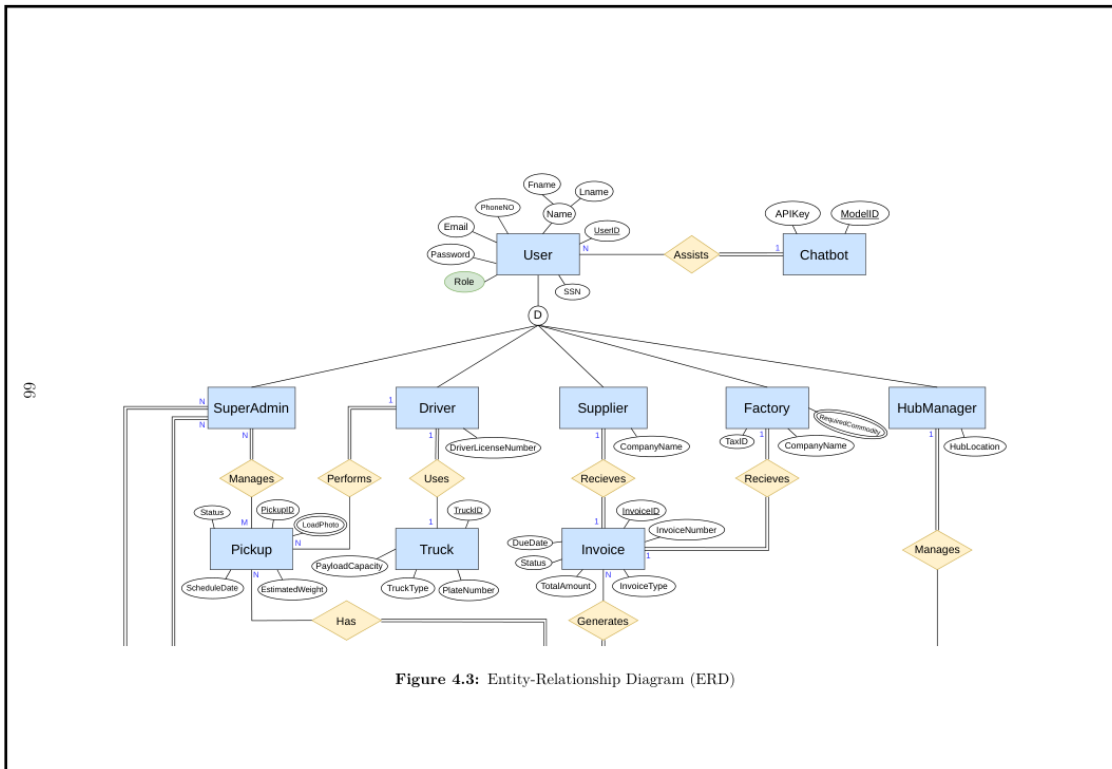


Figure 4.12: Entity-Relationship Diagram - Part 1

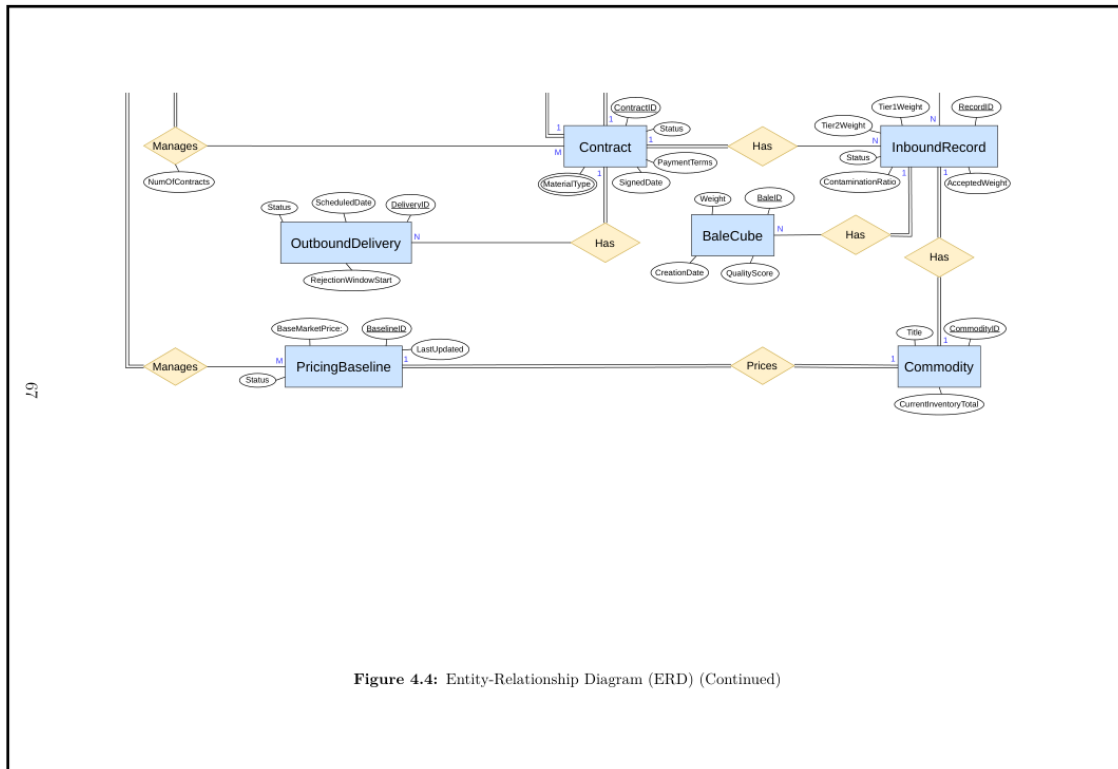


Figure 4.13: Entity-Relationship Diagram - Part 2

4.9 Database Mapping

Database mapping translates ERD entities into relational tables and maps attributes to database columns. Junction tables are used for many-to-many and multi-valued attributes such as material types and pickup photos.

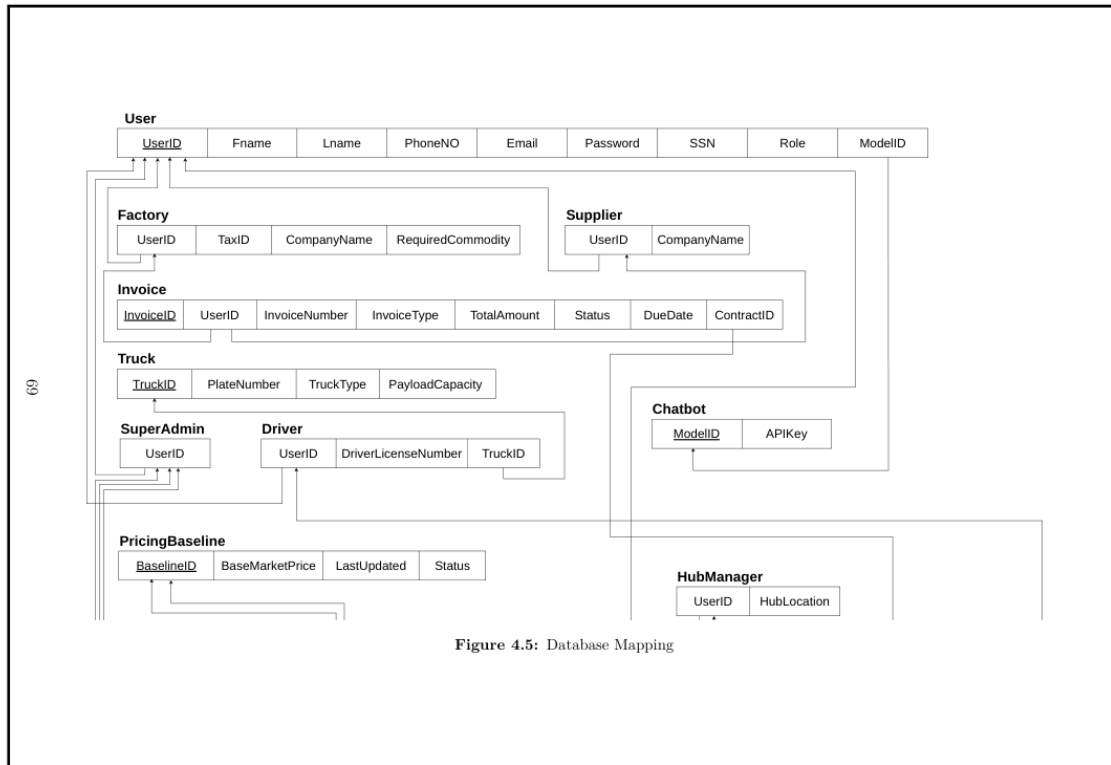


Figure 4.5: Database Mapping

Figure 4.14: Database Mapping - Part 1

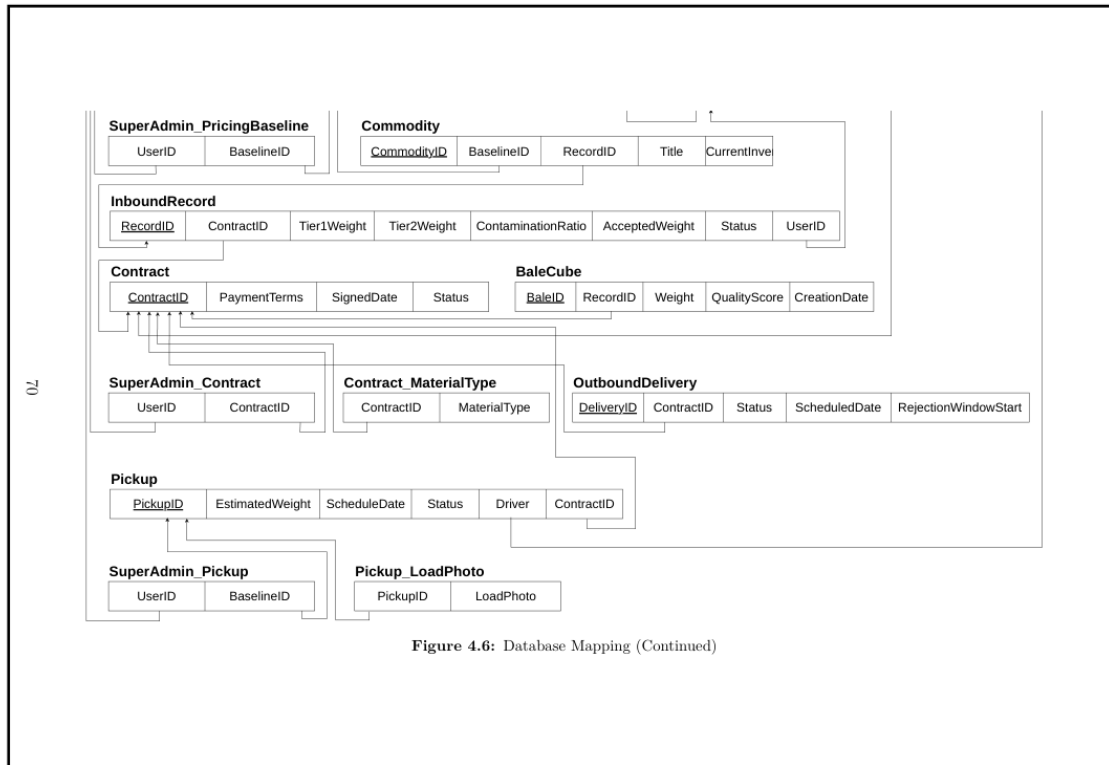


Figure 4.15: Database Mapping - Part 2

4.10 Software Architecture

Layer	Responsibility
Presentation Layer	React dashboards, role navigation, forms, charts, and notifications.
Application Layer	Laravel APIs and services for auth, contracts, pickups, inventory, pricing, invoices, and reports.
Data Layer	SQL database tables for users, contracts, inventory, pricing, and invoices.

Integration Layer	Gemini AI chatbot and future logistics/GPS/notification integrations.
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4.11 UI/UX Guidelines

- Use a clean environmental identity with green/blue accents and modern dashboard cards.
- Keep role-specific navigation simple and only show permitted actions.
- Use popups for registration, material submission, scheduling, contracts, and confirmations.
- Display clear status badges for load, invoice, delivery, and inventory readiness.
- Ensure the interface is responsive for desktop and mobile web users.

4.12 Technology Stack

Technology	Usage
React	Frontend web dashboards and interactive UI.
PHP / Laravel	Backend API, authentication, workflow management, and business logic.
SQL Database	Persistent storage for users, contracts, inventory, pricing, invoices, and operations.
Gemini AI	Onboarding, FAQ, and waste-classification support assistant.
GitHub	Version control, collaboration, and DEPI submission repository.

5. Implementation (Source Code & Execution)

5.1 Source Code Overview

The implementation should be organized into reusable frontend components, backend controllers/services, database migrations, and API routes. The codebase must remain clean, documented, and aligned with the system diagrams.

5.2 Frontend Implementation

- Create role-based dashboards for Supplier, Factory, Driver, Hub Manager, and Super Admin.
- Use forms and popups for registration requests, material data, schedules, contracts, inventory updates, and confirmations.
- Display operational analytics through dashboard cards, charts, tables, and notifications.
- Support responsive web design for desktop and mobile usage.

5.3 Backend Implementation

- Implement authentication, authorization, and role-based permissions.
- Create APIs for users, registration requests, contracts, pickups, trucks, inbound records, bale inventory, pricing baselines, outbound deliveries, and invoices.
- Implement the Algorithmic Pricing Engine: supplier buy price = market baseline x 70%; factory sell price = market baseline x 110%.
- Implement contamination logic: $\text{ratio} = (\text{old weight} - \text{new weight}) / \text{old weight}$; accepted weight is adjusted if contamination exceeds 5%.

5.4 Core Business Rules

Rule	Implementation Meaning
5% contamination threshold	Loads above the threshold trigger deduction or rejection logic and must be logged digitally.
Buy at 70%	Supplier invoice/quotation is calculated at 70% of baseline market price.
Sell at 110%	Factory sales invoice is calculated at 110% of baseline market price.
24-ton FTL threshold	When stock reaches threshold, shipment-readiness alert is sent to the

	Super Admin.
48-hour rejection window	Factory can reject a delivery within 48 hours; otherwise invoice flow continues.
Net 14 / Net 30	Payment due dates are calculated according to contract type and invoice rules.

5.5 Version Control and README

- Upload all source code and documentation to GitHub as required by the DEPI guidelines.
- Use meaningful commits and feature branches for major modules.
- README must include installation steps, system requirements, configuration, execution guide, API documentation, and deployment link if available.

6. Testing & Quality Assurance

6.1 Test Plan

Testing validates that each role can complete its workflow and that the core operational rules are implemented correctly. Testing should include unit testing, integration testing, manual scenario testing, role-access testing, and user acceptance testing.

6.2 Test Cases

ID	Scenario	Expected Result
TC-01	User logs in with valid credentials	User is redirected to the correct role-based dashboard.
TC-02	Supplier submits registration request	Request is saved and visible to Super Admin for review.
TC-03	Super Admin creates pickup schedule	Pickup appears in Supplier and Driver dashboards.
TC-04	Driver confirms supplier arrival and uploads photo	Load status and photo are stored successfully.
TC-05	Hub Manager records Tier 1 and Tier 2 weights	System calculates contamination ratio and accepted weight.
TC-06	Contamination exceeds 5%	Deduction or rejection logic is triggered and logged.
TC-07	Admin updates market	Buy and sell prices

	baseline	update automatically.
TC-08	Commodity stock reaches 24 tons	Shipment-readiness alert is sent to Super Admin.
TC-09	Factory confirms delivery receipt	48-hour rejection window starts and invoice process continues.
TC-10	Unauthorized user opens restricted page	System blocks access or redirects user.

6.3 Bug Reports and Validation

Bug Category	Example	Resolution Approach
Validation issue	Required field accepted as empty	Add frontend and backend validation.
Permission issue	Wrong role accesses restricted dashboard	Fix route guard and authorization middleware.
Calculation issue	Incorrect accepted weight after contamination calculation	Review formula and add automated tests.
UI issue	Table breaks on mobile screen	Improve responsive layout and retest.

User acceptance testing should simulate the complete four-stage workflow: supplier pickup, hub QA, inventory and baling, factory delivery, confirmation, and invoice update.

7. Final Presentation & Reports

7.1 User Manual

User	Manual Summary
Supplier	Log in, view pickups, check weight receipts, view payout history, and use the AI assistant.
Driver	Log in, view daily routes, confirm arrival, upload load photo, and update transit status.
Hub Manager	Confirm hub arrival, enter weights, review contamination, update inventory, and supervise baling.
Factory	View estimated delivery, confirm receipt, view invoices, and track due dates.
Super Admin	Review requests, manage contracts, update market price, create delivery notes, and monitor analytics.

7.2 Technical Documentation

Technical documentation includes the architecture, database schema, role permissions, API endpoints, business rules, deployment steps, and testing results. Diagrams should remain synchronized with the implemented source code.

7.3 Presentation Summary

- Problem: corporate recyclable waste lacks a formal, traceable, and profitable B2B processing channel.
- Solution: contract-based logistics and commodity trading platform with hub QA and factory offtake.
- Innovation: algorithmic pricing, 5% contamination enforcement, 24-ton shipment readiness, ESG support, and AI onboarding.

- Outcome: scalable digital platform supporting circular economy operations and corporate compliance.

7.4 Video Demonstration

A short demonstration video can show login, role dashboards, supplier request, driver route, hub processing, pricing calculation, inventory update, factory confirmation, and final invoice generation.

Appendix A: Business Plan Summary

A.1 Executive Summary

Waste-to-Value generates revenue through commodity arbitrage rather than commission or subscription. Materials are acquired from corporate sources at algorithmically calculated rates and sold as higher-value baled commodities after logistics, QA, sorting, and densification.

A.2 Operational Model

Component	Description
Target Throughput	5 tons/day Year 1 target; approximately 130 to 150 tons/month.
Source Mix	Shopping malls, corporate HQs, hotels, universities, and sports clubs.
Hub Infrastructure	500 sqm aggregation hub, 50-ton vertical hydraulic baler, and 1-ton industrial scale.
Inbound Logistics	1 Jumbo truck and 2 Dababa trucks under dry-lease model.
Outbound Logistics	Subcontracted 24-ton FTL flatbed shipments to factories.

A.3 Pricing and Revenue Model

Pricing Element	Formula / Value
Market baseline	Admin-defined weighted average per commodity.



Supplier buy price	Baseline x 70%.
Factory sell price	Baseline x 110%.
Gross spread	40 percentage points between buy and sell rates.
Payment terms	Net 14 / Net 30 depending on contract type.

Table 2.11: Operational Employee Monthly Salaries (HR Payroll)

Position	Headcount	Monthly Salary (EGP)
Hub Manager	1	25,000
QA Admin	1	12,000
QA Sorters / Ops Staff	4	28,000
Loaders	3	18,000
Drivers	3	24,000
B2B Sales Representative	1	15,000
Gross Staff Payroll	13	122,000
Employer Social Insurance (~15%)		18,300
Total HR Payroll (Staff)		140,300

Table 2.12: Business Operation and Costs

Month	HR Payroll (EGP)	Hub & Tech (EGP)	Logistics (EGP)	Marketing (EGP)	Total / Month (EGP)
January	158,300	53,000	122,000	12,500	345,800
February	158,300	53,000	122,000	12,500	345,800
March	158,300	53,000	122,000	12,500	345,800
April	158,300	53,000	122,000	12,500	345,800
May	158,300	53,000	122,000	12,500	345,800
June	158,300	53,000	122,000	12,500	345,800
July	158,300	53,000	122,000	12,500	345,800
August	158,300	53,000	122,000	12,500	345,800

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Figure A.1: Business Operation and Costs Snapshot

- Monthly Gross Revenue = $130 \times 9,900 = 1,287,000$ EGP.
- Monthly Gross Margin = $1,287,000 - 819,000 = 468,000$ EGP.

2.7.3 Profit Margin

Table 2.14: Profit Margin

Item	Q1 (Jan–Mar)	Q2 (Apr–Jun)	Q3 (Jul–Sep)	Q4 (Oct–Dec)	Total / Year
Total Revenues	3,861,000	3,861,000	3,861,000	3,861,000	15,444,000
Material Acquisition	2,457,000	2,457,000	2,457,000	2,457,000	9,828,000
Gross Margin	1,404,000	1,404,000	1,404,000	1,404,000	5,616,000
Operating Expenses	1,037,400	1,037,400	1,037,400	1,037,400	4,149,600
Profit Before Tax	366,600	366,600	366,600	366,600	1,466,400
Taxes (22.5%)					329,940
Profit After Tax					1,136,460

2.7.4 Cash Flow

Table 2.15: Cash Flow

Item	Pre-Op	Q1 (Jan–Mar)	Q2 (Apr–Jun)	Q3 (Jul–Sep)	Q4 (Oct–Dec)	Total
Cash at Beginning	0	4,149,600	4,516,200	4,882,800	5,249,400	0
Equity Capital	4,707,850	0	0	0	0	4,707,850
Sales Revenue	0	3,861,000	3,861,000	3,861,000	3,861,000	15,444,000
Total Cash In	4,707,850	3,861,000	3,861,000	3,861,000	3,861,000	20,151,850
CAPEX Investment	558,250	0	0	0	0	558,250
COGS + OpEx + Tax	0	3,494,400	3,494,400	3,494,400	3,824,340	14,307,540
Total Cash Out	558,250	3,494,400	3,494,400	3,494,400	3,824,340	14,865,790
Cash at End	4,149,600	4,516,200	4,882,800	5,249,400	5,286,060	5,286,060

Figure A.2: Profit Margin and Cash Flow Snapshot

